

EXPLORATORY DATA ANALYSIS

Dataset 1: Historical Top 200 and Viral 50 (2017–2021)

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1. Overview

This document records the exploratory data analysis (EDA) performed on Dataset 1 — the Kaggle Spotify Historical Charts dataset — as part of the Hidden Gem Music capstone project. EDA was conducted in Power BI using Power Query aggregations across the full cleaned dataset. Due to the dataset's size (~3.5GB, 26 million rows), column-level NULL analysis that exceeded Power BI's memory limits is deferred to SQL Server post-ingestion and noted below.

The purpose of this EDA is to document what was actually observed in the data, flag discrepancies from expected values, and inform ingestion script decisions and stored procedure logic.

2. Dataset Summary

Metric	Observed Value
Total rows (post-cleaning)	26,090,476
Earliest date	January 1, 2017
Latest date	December 31, 2021
Top 200 rows	20,249,193
Viral 50 rows	5,841,283
Unique countries (distinct region values)	70
Maximum credited artists on a single song	4 (artist_primary through artist_4)
NULL streams (estimated)	~5,841,283 — equal to Viral 50 row count; to be confirmed in SQL

3. Chart Type Distribution

Dataset 1 contains two chart types published on overlapping schedules across the same geographic regions and time period.

Chart Type	Row Count	% of Total
Top 200	20,249,193	77.6%
Viral 50	5,841,283	22.4%
Total	26,090,476	100%

The Top 200 accounts for roughly three quarters of all rows. The Viral 50 is a meaningful minority — approximately 5.8 million rows — and is retained in full as it provides the strongest early-stage cross-regional signal for the Hidden Gem scoring logic. Songs appearing on the Viral 50 before crossing into the Top 200 of another country are the most valuable discovery gap indicators in the dataset.

4. Trend Value Distribution

The trend column in Dataset 1 uses categorical string values to describe chart movement. These will be normalised to numeric trend_velocity values at SQL ingestion.

Trend Value	Row Count	Notes
MOVE_DOWN	11,184,303	Normalises to -1.0 at ingestion
MOVE_UP	9,769,681	Normalises to +1.0 at ingestion
SAME_POSITION	3,293,499	Normalises to 0.0 at ingestion — see discrepancy note below
NEW_ENTRY	1,842,993	Normalises to +1.0 at ingestion — strong Hidden Gem signal
Total	26,090,476	

Observed discrepancy: the value used for stationary chart position is SAME_POSITION, not SAME as documented in the project reference materials and database schema ADR. The ingestion script must map SAME_POSITION to trend_velocity = 0.0. If the script is written to map SAME it will fail to normalise approximately 3.3 million rows correctly, leaving them with NULL trend_velocity values.

5. Artist Column Distribution

The artist column was split into up to 4 positional columns during Power BI cleaning. The following counts show how many rows have a credited artist in each position beyond the primary.

Column	Rows with Value	Interpretation
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artist_primary	26,090,476	All rows — every track has at least one credited artist
artist_2	~8,000,000	~30.7% of rows have a second credited artist
artist_3	~2,000,000	~7.7% of rows have a third credited artist
artist_4	~797,000	~3.1% of rows have a fourth credited artist

The maximum observed artist count in Dataset 1 is 4. This is lower than Dataset 2's maximum of 7, which reflects a real shift in music industry crediting conventions between the 2017–2021 and 2023–2025 periods — particularly in Latin, trap, and hyperpop genres where large multi-artist credits became significantly more common in the later period. This is not a data quality issue.

6. Geographic Coverage

Dataset 1 contains chart data for 70 distinct region values. These are stored as full country names (e.g. 'United States') rather than ISO codes. Mapping to the Country lookup table occurs at SQL ingestion via the full_name column.

The complete list of distinct region values and their row counts will be extracted via SQL query post-ingestion as part of the ingestion validation step. Any region values that do not match a full_name in the Country table will require manual review before ingestion is complete.

Dataset 2 covers 73 distinct countries. The 3-country difference is expected — the two datasets have different geographic coverage and there is no requirement that they match.

7. Deferred Observations — Pending SQL Ingestion

The following EDA findings could not be obtained from Power BI due to memory limits on the 26-million-row dataset. Each will be captured via SQL query after BULK INSERT and appended to this document.

Metric	Why Deferred	SQL Query Approach
Exact NULL streams count	COUNTBLANK on streams column exceeded Power BI 1GB query memory limit	SELECT COUNT(*) FROM ChartEntry WHERE streams IS NULL AND source = 'Historical_Top200_and_Viral50'
Exact artist_2/3/4 counts	Estimated from Power BI cards — exact figures to be confirmed	SELECT COUNT(*) FROM ArtistSong JOIN Song — count songs with 2, 3, 4 ArtistSong rows
spotify_id parse success rate	Not measurable until URL parsing runs at ingestion	SELECT COUNT(*) FROM Song WHERE spotify_id IS NULL AND song_id IN (DS1 songs)
Country name match rate	Not measurable until ingestion runs against Country table	SELECT region FROM staging WHERE region NOT IN (SELECT full_name FROM Country)

Exact distinct region count breakdown	Table visual with 70 rows would have required scrolling; full list captured in SQL	SELECT region, COUNT(*) FROM ChartEntry GROUP BY region ORDER BY region
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8. Key Findings for Ingestion and Stored Procedure Authors

1. SAME_POSITION, not SAME — the trend column uses SAME_POSITION as the stationary value. The ingestion script must map this correctly to trend_velocity = 0.0.
2. NULL streams is by design — all 5,841,283 Viral 50 rows are expected to have NULL streams. This is a documented dataset behavior, not a data quality error. The ingestion script must not reject or attempt to fill these rows.
3. NEW_ENTRY is a signal, not zero — 1,842,993 rows carry the NEW_ENTRY trend value. This normalises to +1.0 in trend_velocity, not 0.0. A song's first appearance in a chart is the strongest Hidden Gem indicator and must be treated as a positive movement value.
4. Date range confirmed — January 1, 2017 through December 31, 2021. This matches the documented dataset coverage. The 22-month gap to Dataset 2 (which begins October 18, 2023) is confirmed.
5. 70 distinct countries — the Country lookup table must contain matching full_name entries for all 70 region values observed. Any unmatched values will cause FK violations at ingestion.